

Homework Assignment 2

Sensorimotor Neurotechnology 2025

Exploring Encoding Models

In Homework 2, we will become familiar with the proportional and biomimetic encoding models used in Valle et al., 2018 *Neuron*. The goal of the homework is twofold: First, to understand how each model encodes stimulus features in the stimulation current (via modulations in amplitude or frequency); and Second, to understand how the different parameters of each encoding model shapes the resulting stimulation.

Encoding models

Below are the four encoding models used by Valle and colleagues. The equations may not be the same in some cases, but they capture the essence of the encoding model. Use these equations for the homework.

Amplitude encoding model (ANM)

The ANM encoding model translates the sensor readings into a proportional current amplitude.

$$A = A_{min} + (A_{max} - A_{min}) \left(\frac{s - s_{15}}{s_{75} - s_{15}} \right)$$

Where s is the current sensor reading and s_N is N % of the maximum sensor reading. A_{max} and A_{min} are the maximum and minimum current amplitudes. In practice, these are set based upon the necessary threshold for perceiving a sensation (A_{min}) and the max non-painful amplitude (A_{max}). When $s > s_{75}$, then $A = A_{max}$. When $s < s_{15}$ then $A = 0$. The pulse frequency F of the current is fixed at 50 Hz.

Frequency encoding model (FNM)

The FNM encoding model is the first biomimetic encoding model from Valle 2018. It takes the millisecond-by-millisecond spiking output of the skin-neuron model *TouchSim* and converts it into a pulse frequency F proportional to the firing rate of the simulated mechanoreceptors.

The steps for this are as follows:

- 1) The sensor readings from a prosthetic finger are translated into an estimated spatio-temporal pattern of skin indentations.
- 2) This pattern serves as the stimulus input for simulated mechanoreceptor populations.
- 3) The input is converted into millisecond spike timings for each mechanoreceptor in each population.
- 4) The activity of each mechanoreceptor is summed to create a vector of population spike timings, R .
- 5) The population activity R is converted into binned firing rates, R_{bin} . This is done using 10 ms bin widths and taking the average R within the bin window. Note that R_{bin} is a rate, so it needs to be converted into units Hz.

Finally, we encode the binned population activity into pulse frequency F as follows:

$$F = F_{min} + (F_{max} - F_{min}) \left(\frac{R_{bin}}{R_{max}} \right)$$

Where R_{bin} is the binned population firing rate, R_{max} is the maximum population firing rate, and F_{max} and F_{min} are the maximum and minimum pulse frequencies. When $R_{bin} > R_{max}$ then $F = F_{max}$. The amplitude A is fixed at the value midway between the measured minimum and maximum (see above).

Note that in the original formulation by Valle and colleagues, F was set to be equal to R_{bin} . However, this can be problematic as the population firing rate is tied to how many mechanoreceptors are modelled in the population, and the frequency can fall outside the temporal resolution of mechanoreceptors and what is perceivable (~ 1000 Hz). I therefore prefer the above formulation, which provides a bit more flexibility.

For the homework, I have already completed steps one through three. You will only need to complete steps four and five.

Hybrid encoding model 1 (HNM-1)

In this encoding model, the current amplitude A is determined using ANM and the pulse frequency F is determined using FNM.

Hybrid encoding model 2 (HNM-2)

In this encoding model, the pulse frequency F is determined using FNM and the current amplitude A is based on the number of activated nerve fibers.

$$A = A_{min} + (A_{max} - A_{min}) \left(\frac{N_{act}}{N_{total}} \right)$$

Where N_{act} is the number of activated mechanoreceptors within the current bin, and N_{total} is the number of mechanoreceptors in the entire population.

Homework Assignment

The purpose of this assignment is to give you experience with different encoding models for restoring touch to a prosthesis. You will do so using simulated sensor readings and the outputs of the TouchSim skin-neuron model. The description of the encoding models and the accompanying equations above should be sufficient to recreate the encoding models used by Valle and colleagues.

I have included a .mat file (HW2_EncodingModels.mat) with the simulated sensor readings for two types of stimuli (ramping and stepping) and the corresponding millisecond-by-millisecond spiking of the simulated mechanoreceptor populations (we will focus on SA1 and RA) that has been output from TouchSim.

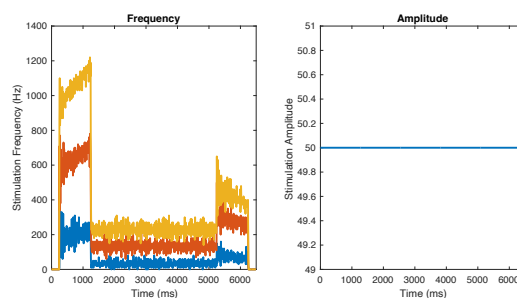
The relevant matrices are as follows.

Stimuli. The ramping stimuli are stored in the matrix *Stim1* and the stepping stimuli are stored in the matrix *Stim2*. The dimensions of each are 3x6500, where each row corresponds to a different level of force and each column corresponds to a single millisecond. The values provided correspond to s_N .

TouchSim. I have simulated the spiking of 50 SA1 and 50 RA mechanoreceptors to each stimulus type. The spiking of each population to each stimulus is stored in a matrix whose name corresponds to the mechanoreceptor and stimulus type. For example, the RA population's responses to the ramping stimuli are stored in *RA_Spikes1* and to the stepping stimuli in *RA_Spikes2*. The dimensions of each are 50x6500x3 with the contents corresponding to *mechanoreceptor* x *time* x *force*. Therefore, the element at position (32,200,2) would correspond to the 32nd mechanoreceptor's response at the 200th millisecond to the second force level.

Below I include six questions aimed at helping you get an intuition for the different decoding models and their parameters. If you are working in a group, you need to answer all six questions. If you are working alone, you only need to answer Questions 1–4. For an extra point, you can answer one of the additional two questions.

Question 1. Given the ramping stimuli, translate the sensor reading and TouchSim responses into stimulation parameters for each of the four encoding models. Use the following parameters for the encoding models: $A_{min} = 5$; $A_{max} = 20$; $F_{min} = 0$; $F_{max} = 1000$; $R_{max} = 1000$. For each encoding model, plot how the stimulation amplitude and pulse frequencies vary over time. I recommend using the below plotting format where each force value is a differently colored line:



Question 2. Given the stepping stimuli, translate the sensor reading and TouchSim responses into stimulation parameters for each of the four encoding models. Use the same parameter values as for Question 1. As before, for each encoding model, plot how the stimulation amplitude and pulse frequencies vary over time.

Question 3. For each of the four encoding models explain in a few sentences how the stimulation amplitude and pulse frequencies encode the static and dynamic aspects of the stimulus.

Question 4. Let's explore how the stimulation parameters are affected by the choice of parameter ranges during encoding. We will focus on the encoding of the stepping stimuli (high force value) by ANM and FNM.

For ANM: Plot how the stimulation amplitude changes with the following three s_N value pairs: (i) s_{10} and s_{90} ; (ii) s_{40} and s_{70} ; and (iii) s_{20} and s_{60} . Explain in a few sentences what stimulus information is lost and gained by these choices for parameter values.

For FNM: Plot how the stimulation amplitude changes with the following three R_{max} values: 2000, 500, 250. Explain in a few sentences what stimulus information is lost and gained by these choices for parameter values.

Question 5. Let's explore how the biomimetic encoding models (FNM, HNM-1, and HNM-2) change when using individual mechanoreceptor populations. Redo Question 2, but only include SA1 or RA as the encoded population. For each mechanoreceptor type, set $F_{max} = 500$; and $R_{max} = 500$. How does this change what information is encoded? How does this depend on the mechanoreceptor type used? Any insights into to why RA-only might feel more natural than SA1-only (see, Valle's results)?

Question 6: Plot each encoding model's stimulation (amplitude and frequency) in the format of Figure 1A (rightmost portion) from Valle et al., (2018). Do so for the medium force-level of the stepping frequency, using the encoding parameters from Q1.